

RS minimal model: floor summary and literature reproductions

working notes, June 2026. Quark sector, minimal (non-custodial) Randall-Sundrum.

Quick notes after the constraint-code audit and fixes. Two things here: (1) the floor summary (what M_{KK} the minimal model actually needs), and (2) our reproductions of the relevant published figures, showing just our own panel in each case. All run on the fixed code.

1. Floor summary

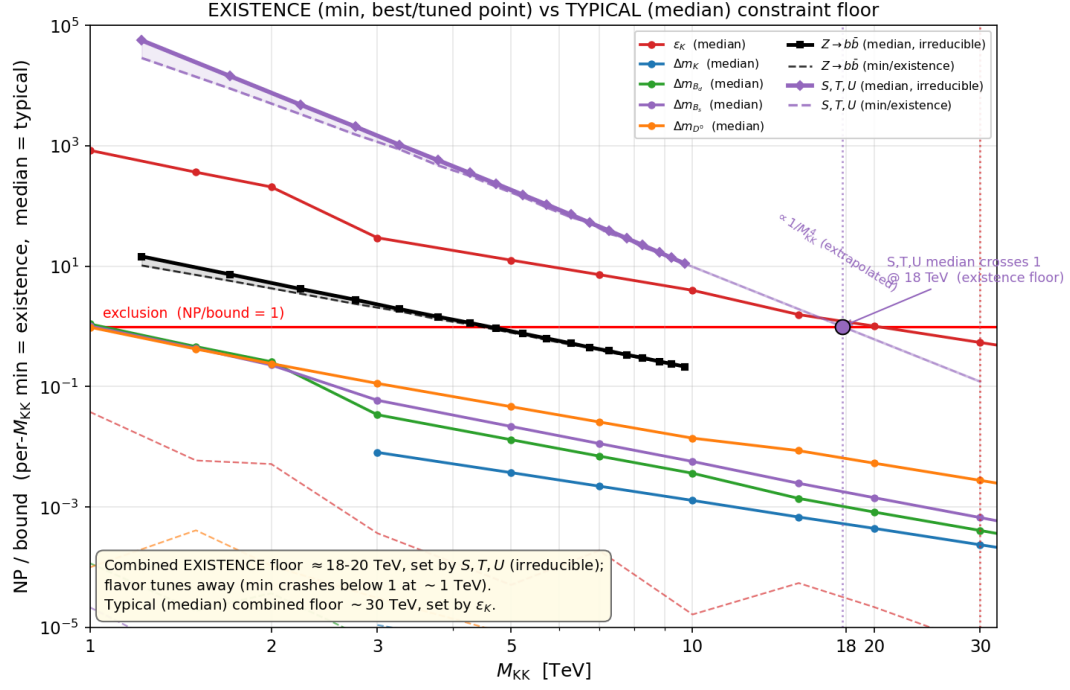
There are two honest notions of a floor, and they are very different:

- **Typical floor:** where the *median* anarchic point becomes allowed. This is the usual literature statement. With current data it is about **30 TeV**, set by ε_K .
- **Existence floor:** where even a single best-case (fine-tuned) point is allowed. This is about **18 to 20 TeV**, set by S, T, U .

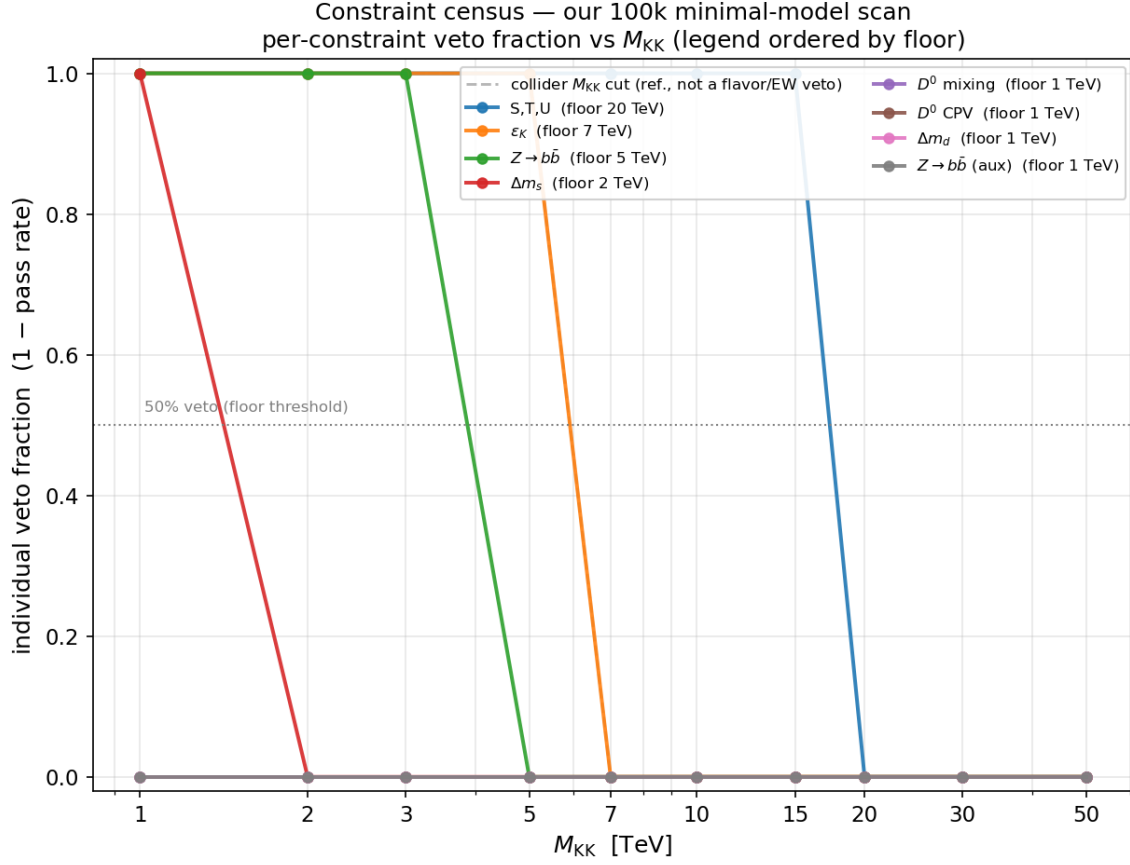
The split matters because the flavor constraints are tunable (you can align phases so the new-physics piece cancels, e.g. $\varepsilon_K \propto \text{Im } M_{12} \rightarrow 0$), but S, T, U is not: in minimal RS the oblique T depends only on M_{KK} and the warp volume, with no Yukawa freedom. So even a fine-tuned minimal RS point cannot sit below about 18 to 20 TeV, and that wall is the oblique S, T, U bound. This is exactly why custodial RS exists (it protects T).

Constraint	Typical (median) floor	Existence (best/tuned) floor	Tunable?
ε_K	~ 30 TeV (current)	below ~ 1 TeV	yes (align $\text{Im } M_{12} \rightarrow 0$)
$\Delta m_d (B_d)$	few TeV	below ~ 1 TeV	yes
$\Delta m_s (B_s)$	few TeV	below ~ 1 TeV	yes
D^0	low	below ~ 1 TeV	yes
Δm_K	low	below ~ 3 TeV	yes
$Z \rightarrow b\bar{b}$ (T010)	~ 5 TeV	~ 4.6 TeV	no (gauge, c_{Q3} set by m_t)
S, T, U (EW001)	~ 20 TeV	~ 18 to 20 TeV	no (no Yukawa dependence)
Combined	~ 30 TeV (ε_K)	~ 18 to 20 TeV (S, T, U)	

Existence vs typical, per constraint. Min ratio-to-bound (best, fine-tuned point) and median ratio (typical point) vs M_{KK} . The flavor curves crash well below 1 (tunable, no real floor); S, T, U has min $\approx$ median and crosses 1 only near 18 to 20 TeV (irreducible).



Our 100k scan, constraint census. Per-constraint veto fraction vs M_{KK} from our minimal scan with all fixes in. Ranking: S, T, U at 20, ε_K at 7, $Z \rightarrow b\bar{b}$ at 5 (this was the old 25 to 30 TeV artifact before the fix), Δm_s at 2. Dropping $Z \rightarrow b\bar{b}$ does not move the combined floor, so it is not the driver.

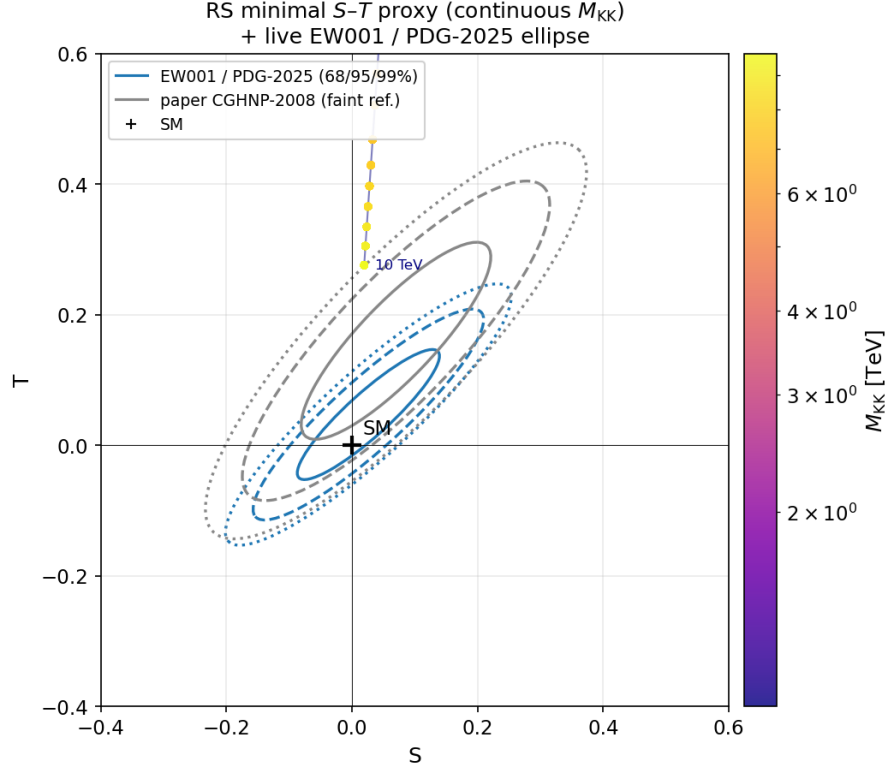


2. Reproductions (our panel only)

For each, this is just our reproduction. The real published figure is in the longer report; here we show only the thing we are actually reproducing.

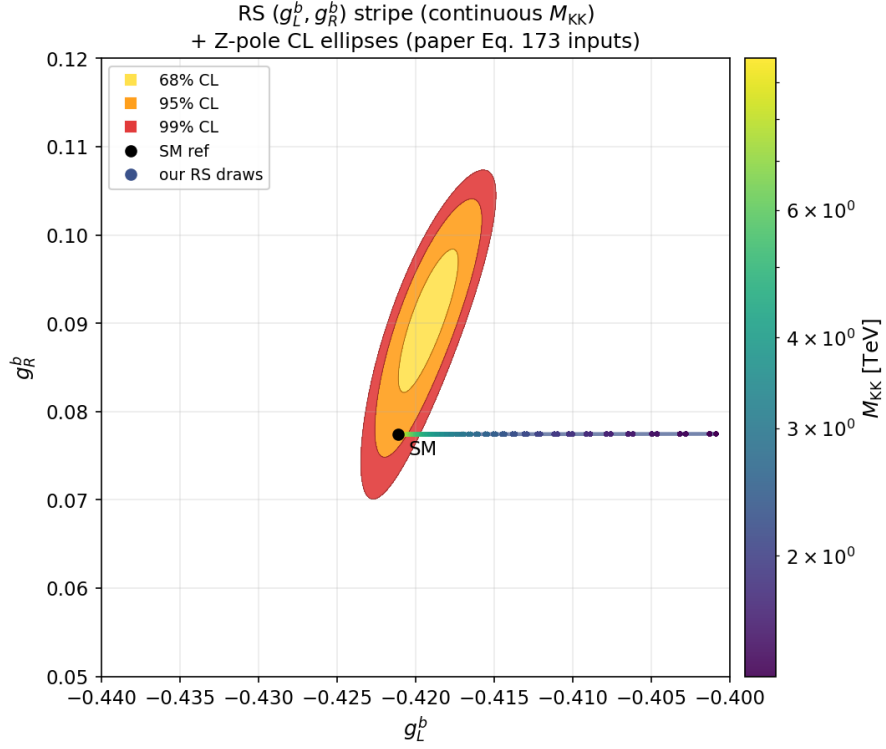
S, T, U oblique (vs Casagrande et al. 0807.4937, Fig. 4)

Our S - T plane. The RS minimal wedge climbs steeply in T (the volume-enhanced T problem) and leaves our PDG-2025 ellipse fast; grey is the 2008 ellipse for reference. This is the irreducible bound that sets the existence floor.



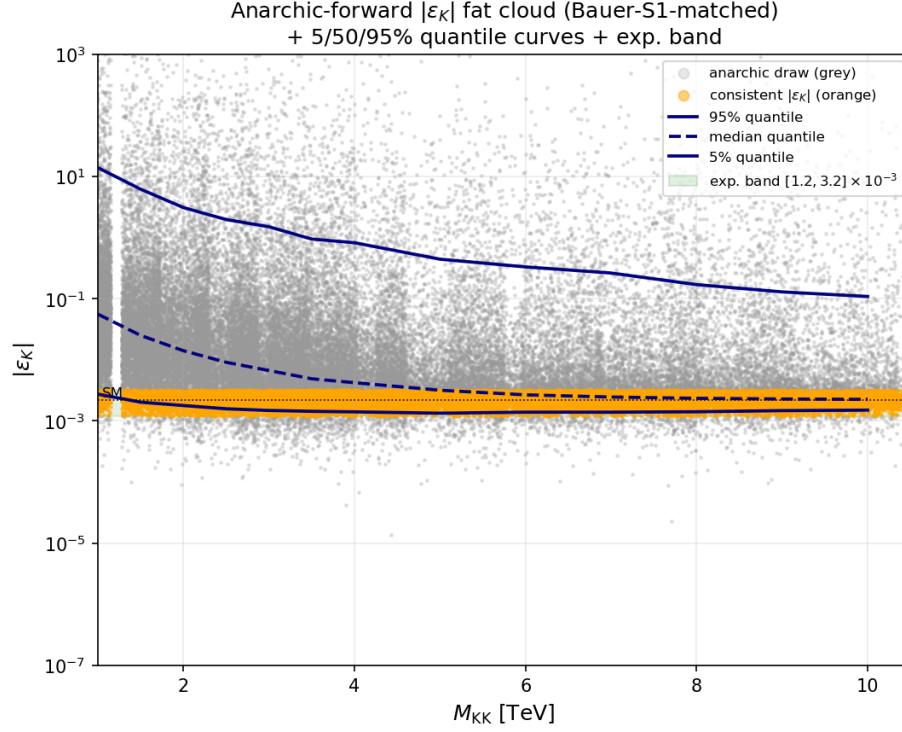
$Z \rightarrow b\bar{b}$ couplings (vs Casagrande et al. 0807.4937, Fig. 8)

Our (g_L^b, g_R^b) plane. The RS stripe pins at $g_R^b \approx 0.077$ and slides to less-negative g_L^b as M_{KK} drops, staying right by the SM point. With the B1 fix this is gauge-dominated and weak (bites only to about 5 TeV), not the 25 to 30 TeV the bug produced.



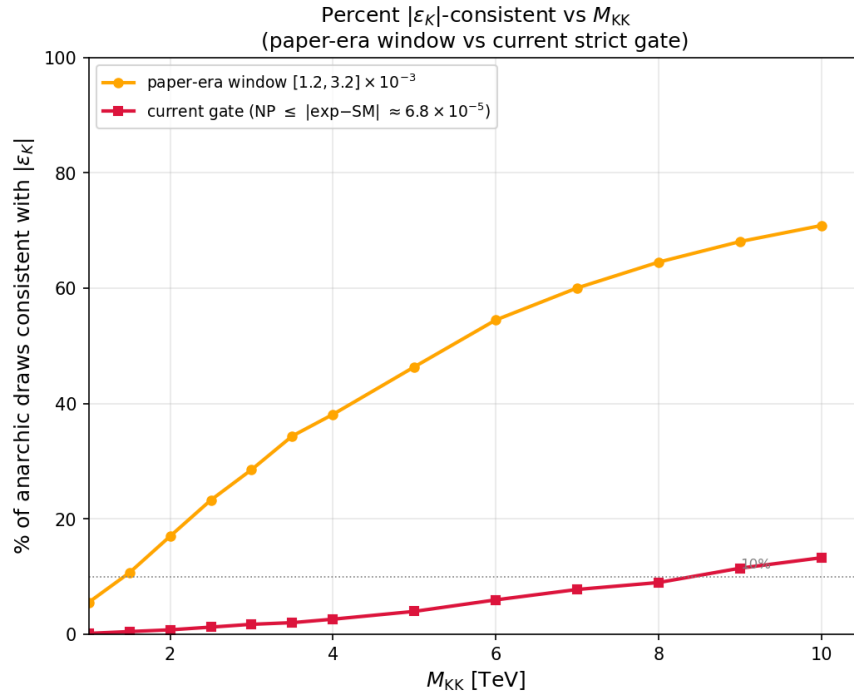
ε_K (vs Bauer et al. 0912.1625, Fig. 4)

Our anarchic $|\varepsilon_K|$ cloud, using Bauer's S1 draw setup ($Y_{\max} = 3$, scanned c). Fat, about 6 to 7 decades, with the 5/50/95 percent quantile curves. The 95 percent quantile enters the allowed band near 10 TeV, which is Bauer's headline number.



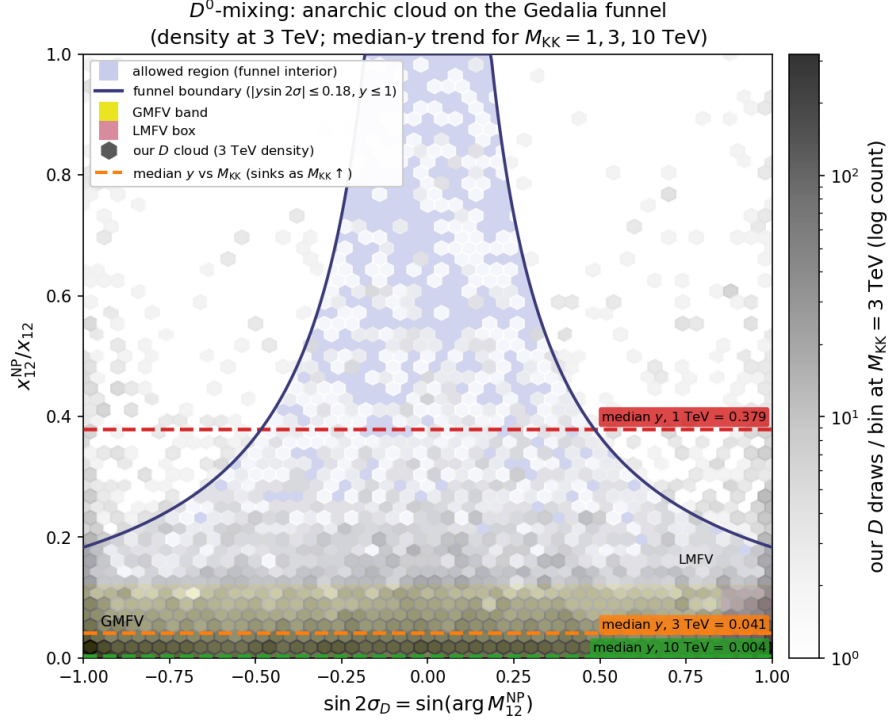
Consistency fraction (vs Bauer et al. 0912.1625, Fig. 5)

Fraction of anarchic points consistent vs M_{KK} , paper-era window vs current strict gate. The rising $1/M_{KK}^2$ decoupling is the same shape as the paper.



D^0 mixing (vs Gedalia, Grossman, Nir, Perez 0906.1879, Fig. 1)

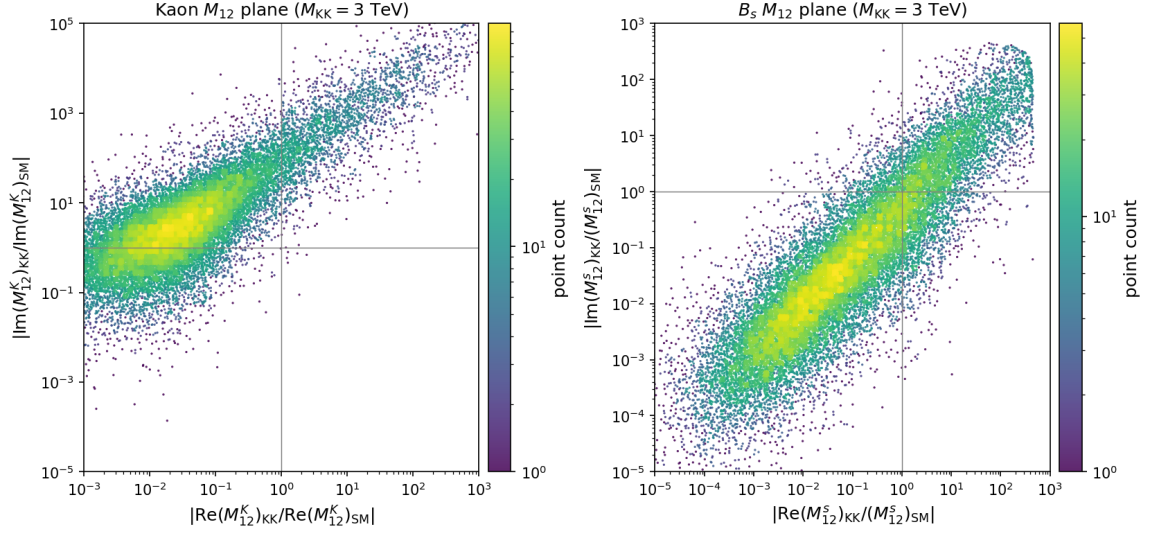
Our anarchic D -meson cloud on the Gedalia funnel, cleaned up to a single 3 TeV density plus the median- y trend at 1, 3, 10 TeV. The cloud sinks into the allowed funnel as M_{KK} rises. (Optional figure: say the word if you would rather drop it.)



$\Delta F = 2$ amplitudes (vs Blanke et al. 0809.1073, Fig. 2)

Re vs Im of the KK contribution to M_{12} , for K (left) and B_s (right). For kaons the imaginary part dominates the real part by about $100\times$ (the ε_K problem); ours is $106\times$.

NP (KK) contribution to M_{12} , normalized to SM — kaon (left), B_s (right)



Dropped on purpose: the B_d and B_s phase planes (the old Figs 6 and 7). Our model clouds are fine, but the experimental CL contours we drew there are only Gaussian approximations of the CKMfitter regions, not the exact published contours, so they are not a clean reproduction. Left out rather than shown as if exact.